Responsible Innovation Statement

In this section, we reflect on the ethical aspects of the AI-powered educational application, with reference to the core concepts of opacity, moral deskilling, and human-centred AI. We identify the societal and ethical questions raised by our design and propose specific recommendations to ensure the innovation is developed responsibly, aligning with the values of our end-users: children, parents, and educators.

1.1 Opacity

The use of a Large Language Model (LLM) as the core of our application introduces the challenge of opacity, often referred to as the black box [12] problem. It is not inherently clear how the LLM generates a specific scenario or formulates its feedback. This lack of transparency raises significant ethical questions regarding trust. If the AI produces a biased scenario or provides pedagogically unsound advice, who is responsible? Furthermore, for educators and parents to trust this tool, they must have confidence that its outputs are based on sound ethical and developmental principles, not on statistical artifacts from its training data.

To mitigate these risks, we will implement the following design patterns:

1. **Justification Based Feedback [11]:** Instead of simply stating a choice is "good," the AI's feedback mechanism will be explicitly prompted to justify its response by referencing the core ethical concept at play. For example, it will generate feedback like, "Choosing to tell the teacher was a brave decision that shows you value **fairness**," or "This response shows great **care** for your friend's feelings." This makes the underlying reasoning transparent to the user.
2. **Parameter Transparency for Parents/Teachers:** A dashboard feature will allow parents or edu-cators to see the parameters that shaped a session. It could display the ethical frameworks (e.g., Virtue Ethics, Ethics of Care) and specific topics (e.g., Cyberbullying, Social Exclusion) covered, providing insight into the educational content without needing to understand the model's technical workings.
3. **Human Validation:** A critical component of our development process will be a review panel of child psychologists and experienced educators. This panel will lead to creating a "Gold Standard"[3] dataset for fine-tuning, ensuring that the model's outputs are aligned with expert consensus.

1.2 Cultural Appropriateness & Bias

There is a concern in LLMs and other AI that could pose a problem when generating ethics cases. It relates to the inherent bias that an AI system has when it has been trained on data that has been selected by humans [15]. How this relates to cultural appropriateness comes from the fact that the datasets used to train such systems often reflect the dominant cultural, social, and moral perspectives present in the contexts from

which the data were drawn. As a result, the generated cases may unintentionally prioritize Western ethical frameworks or exclude culturally diverse moral reasoning traditions. This becomes particularly relevant when ethical cases are generated about sensitive topics such as bullying for use with children in primary education.

If an Ethics Case Generator produces bullying scenarios that only reflect certain cultural or social norms (such as those common in Western classroom settings), it might overlook important variations in how bullying is expressed, perceived [14], or addressed in other cultural contexts [5]. For example, in some cultures, indirect forms of social exclusion are more prevalent, while in others, verbal teasing may be viewed differently in terms of severity or intent. Without careful oversight, the generated cases might inadvertently reproduce stereotypes about gender, ethnicity, or family background, or fail to represent the diversity of experiences and values found in real classrooms [13].

In addition, children's moral understanding develops through social and cultural interaction, which means that ethics education must connect to their lived experiences and community values [21]. If the generated cases do not align with the moral frameworks familiar to the children's (think norms around respect, em-pathy, and cooperation) they risk being misunderstood or being culturally alienated. For this reason, it is crucial that any Ethics Case Generator designed for primary education incorporates diverse, age-appropriate examples of bullying and moral conflict that resonate with the children's own school and cultural environments. To achieve this, human educators and researchers should remain closely involved in the selection, review and adaptation of AI-generated cases to ensure cultural sensitivity and pedagogical relevance.

1.3 Unpredictability in System Performance

A significant challenge is the unpredictability of child as end-users, who often provide unexpected, off-topic inputs. This poses a direct risk to the LLM's robustness, potentially causing system failure or inappropriate pedagogical responses. To mitigate this, our Validation Layer (2.3.2) and dedicated Safety and Filtering Module (2.3.1) are designed to handle such unexpected requests. If a child's input is unrelated to the scenario, the system will not attempt to generate a response; instead, it will deploy a safe reply from pre-defined context, to gently redirect the user back to the task, ensuring both system stability and educational integrity.

1.4 Data privacy, Accessibility and Platform Integration

Protecting our minor end-users is the most important, requiring strict GDPR compliance [4]. All student data will be anonymized by default before display on the teacher dashboard. This permits trend analysis while ensuring individual privacy. Data will never be used to train external, third-party LLMs.

The "Text + Audio Mode" is a foundational component of the application's commitment to accessibility and inclusive design. These modes are essential for ensuring equitable access for all learners. For children with visual impairments, this feature provides full audio narration of all content, including the generated scenarios, ethical questions, and the AI's feedback, allowing them to participate fully without relying on visual text.

For platform integration, the tool will be implemented as a standard, secure web application to maximize compatibility and minimize the technical adaptive cost for schools.

1.5 Reflection on Design Journey and Future Impact

The pivotal decision to use a generative LLM instead of a static database of cases (3) prioritizes dynamic, customizable content. This choice dictated our "human-in-the-loop" and "safety-first" approach, realized in our modular architecture (2.3.1). This design embeds safety filters and is guided by a pedagogical commitment to reflective practice [16], shaping our feedback module to prioritize justification [10] over simple right/wrong answers.

The immediate impact is to shift classrooms from rule-recitation to active moral exploration (2.2). In the long term, the "EthiCase Kids" framework is highly scalable and can be expanded to cover other dilemmas like digital citizenship or academic honesty. Furthermore, the expert-

validated ”Gold Standard” dataset [3] will become a valuable asset for benchmarking child-AI moral alignment [2].